

Analysis Notes

- BR = bark roughness
- CT = bark condensed tannins
- CN = bark carbon-nitrogen ratio
- PC = percent cover (percent occupied cells in the sampling grid)
- SR = lichen species richness
- L = size of lichen network (number of nodes)
- Cen = centrality of the lichen network

```
## Make sure to run make.R first and load the necessary objects.
```

```
library(pacman)
p_load(drake, xtable, rmarkdown)
loadadd(onc.dat)
loadadd(cn.d.onc)
```

Is the network similarity driven by abundance of lichen?

We found support for the hypothesis that genotypically based variation in lichen network structure is potentially driven by variation in bark roughness. In an initial PerMANOVA with sequential partitioning of variance explained, genotype has a strong, statistically significant effect ($R^2 = 0.37$, $p\text{-value} = 0.040$) when including tree traits and lichen community indices in the model. In this ordering, condensed tannins ($R^2 = 0.09$, $p\text{-value} = 0.486$) and species richness ($R^2 = 0.11$, $p\text{-value} = 0.017$) are both significant, although with low explanatory power. After moving genotype to the final predictor in the model, genotype is no longer significant and has low explanatory power, while both bark roughness ($R^2 = 0.14$, $p\text{-value} = 0.006$) and species richness ($R^2 = 0.20$, $p\text{-value} = 0.002$) are now significant and have increased in the variance in network similarity explained. Taken together, these results suggest that there is potentially a causal relationship between genotypically explained variance in bark roughness and lichen community species richness that affects the similarity of lichen network structure.

```
cn.geno.trait.pc.sr <- vegan::adonis2(
  cn.d.onc ~ geno + CT + pH + CN + BR + PC + SR,
  by = "term",
  data = onc.dat,
  mrank = TRUE,
  permutations = 100000)

cn.trait.pc.sr.geno <- vegan::adonis2(
  cn.d.onc ~ CT + pH + CN + BR + PC + SR + geno,
  by = "term",
  data = onc.dat,
  mrank = TRUE,
  permutations = 100000)
```

```
as.data.frame(cn.geno.trait.pc.sr)
```

```
as.data.frame(cn.trait.pc.sr.geno)
```

```
xtable(as.data.frame(cn.geno.trait.pc.sr), digits = digits)
```

```
% latex table generated in R 4.5.3 by xtable 1.8-8 package % Mon Jun 8 19:10:04 2026
```

```
xtable(as.data.frame(cn.trait.pc.sr.geno), digits = digits)
```

```
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```

	Df	SumOfSqs	R2	F	Pr(>F)
geno	9	257.29	0.37	2.35	0.040
CT	1	60.20	0.09	4.95	0.048
pH	1	7.82	0.01	0.64	0.445
CN	1	13.75	0.02	1.13	0.297
BR	1	14.72	0.02	1.21	0.282
PC	1	11.13	0.02	0.92	0.353
SR	1	73.32	0.11	6.03	0.016
Residual	21	255.22	0.37		
Total	36	693.44	1.00		

	Df	SumOfSqs	R2	F	Pr(>F)
CT	1	42.57	0.06	3.50	0.073
pH	1	10.00	0.01	0.82	0.373
CN	1	19.52	0.03	1.61	0.210
BR	1	99.98	0.14	8.23	0.006
PC	1	29.88	0.04	2.46	0.122
SR	1	140.15	0.20	11.53	0.002
geno	9	96.14	0.14	0.88	0.564
Residual	21	255.22	0.37		
Total	36	693.44	1.00		

How does species richness relate to network structure?

Examination of the variaince in lichen network similarity explained by species richness and the network metrics of size and centrality, supported the hypothesis that network structure is both a function of increasing numbers of species (i.e., nodes in the network) and other structural differences. In analyzing two PerMANOVAs with these factors, we found that network size ($R^2 = 0.77$, p-value < 0.001) and centrality ($R^2 = 0.04$, p-value < 0.001) both explained vaiation in network similarity regardless of whether species richness was the first or last factor entered into the model. Additionally, although variation in network similarity explained by centrality was low, it was significant even after partitioning the variance explained by network size. Based on this, we can conclude that network similarity is both of function of the number of lichen species in the community determining network size and other aspects of network strcutre, such as centralization.

```
cn.l.cen.sr <- vegan::adonis2(
  cn.d.onc ~ L + Cen + SR,
  by = "term",
  data = onc.dat,
  mrank = TRUE,
  permutations = 100000)

cn.sr.l.cen <- vegan::adonis2(
  cn.d.onc ~ SR + L + Cen,
  by = "term",
  data = onc.dat,
  mrank = TRUE,
  permutations = 100000)

as.data.frame(cn.l.cen.sr)

as.data.frame(cn.sr.l.cen)

xtable(as.data.frame(cn.l.cen.sr), digits = digits)
```

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	Df	SumOfSqs	R2	F	Pr(>F)
L	1	601.44	0.87	376.37	0.000
Cen	1	36.58	0.05	22.89	0.000
SR	1	2.69	0.00	1.68	0.196
Residual	33	52.73	0.08		
Total	36	693.44	1.00		

```
xtable(as.data.frame(cn.sr.l.cen), digits = digits)
```

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	Df	SumOfSqs	R2	F	Pr(>F)
SR	1	77.83	0.11	48.71	0.000
L	1	536.36	0.77	335.64	0.000
Cen	1	26.52	0.04	16.60	0.000
Residual	33	52.73	0.08		
Total	36	693.44	1.00		